



Yuyang Chen

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LinkedIn

Education

ShanghaiTech University, Shanghai, China
School of Information Science and Technology
Ph.D. Student | Advisor: Prof. Hao Geng

2022 - 2027 (Expected)

Research interests: Design for manufacturability (DFM), computational lithography, AI for EDA, and foundation models for semiconductors.

Profile

Ph.D. student researching DFM and computational lithography, with a focus on applying large language models, large vision models, Transformers, and diffusion models to EDA. Research spans hotspot detection, full-chip mask optimization, inverse lithography, and 3D aerial image simulation. Experienced in problem formulation, dataset construction, model design, differentiable simulation, experimental evaluation, and academic writing.

Selected Research

Hotspot Detection

LLM-HD: Layout Language Model for Hotspot Detection with GDS Semantic Encoding

DAC 2024 | First Author

Proposed a hotspot detection method combining GDS semantic encoding with language models to avoid layout information loss from image-based inputs and improve detection performance.

HyDAS: Hybrid Domain Deformed Attention for Selective Hotspot Detection

IEEE TCAD | First Author

Proposed a framework integrating hybrid-domain inputs, deformed attention, and selective learning to improve detection on complex layouts and support reliable deployment.

Mask Optimization and Inverse Lithography

LVM-MO: A Large Vision Model Pioneer on Full-Chip Mask Optimization

DAC 2025 | Co-first Author

Developed a large vision model framework for full-chip mask optimization, significantly improving optimization accuracy and inference efficiency at full-chip scale.

TokMan: Tokenize Manhattan Mask Optimization for Inverse Lithography

NeurIPS 2025 | Co-first Author

Formulated Manhattan mask optimization as discrete sequence generation, using a Diffusion Transformer to balance manufacturability and optimization quality in inverse lithography.

High-Fidelity Lithography Simulation

N3Litho: A GPU-Native Numerical-Neural Framework for 3D Mask Modeling, 3D Propagation, and 3D Aerial Image Simulation

DAC 2026 | First Author

Proposed a GPU-native numerical-neural framework for 3D lithography simulation, enabling accurate and efficient 3D aerial image simulation for advanced technology nodes.

Publications and Honors

- First-author or co-first-author research on **LLM-HD, HyDAS, LVM-MO, TokMan, and N3Litho** in AI for EDA and computational lithography.
- **Best Paper Award Nominations: DAC 2024 and DAC 2025.**

Skills

- **Research:** Layout/GDS representation, foundation models for EDA, physics-aware modeling, full-chip optimization, and GPU-native simulation.
- **Tools and environments:** Python, Linux, $\text{\LaTeX}$.